INCH-POUND

MS25323P
24 September 2024
SUPERSEDING
MS25323N
12 March 2024

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, SOCKET MOUNTED, HERMETICALLY SEALED

Inactive for new design after 5 June 1987.
For new designs use MIL-PRF-83536/9, /10, or /11.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

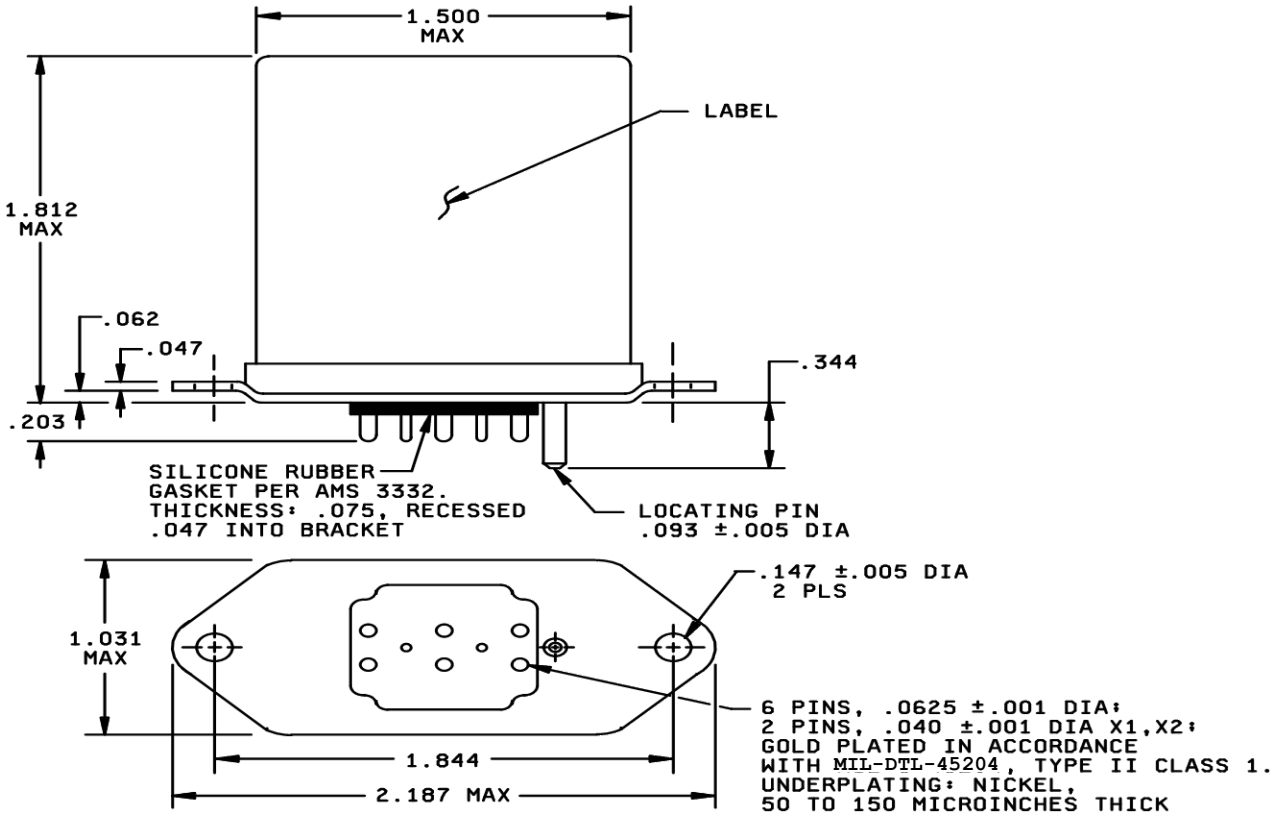
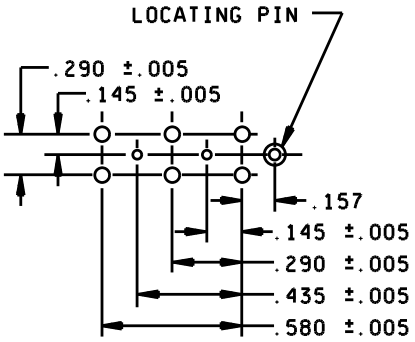


FIGURE 1. Dimensions and configuration.

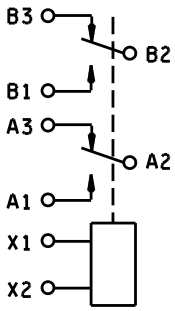
AMSC N/A
FSC 5945

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PIN LAYOUT



CIRCUIT DIAGRAM
SEE NOTES 4 AND 5

Inches	mm	Inches	mm	Inches	mm
.0010	0.025	.093	2.36	.435	11.05
.001	0.03	.145	3.68	.580	14.73
.005	0.13	.147	3.73	1.031	26.19
.040	1.02	.157	3.99	1.500	38.10
.047	1.19	.203	5.16	1.812	46.02
.062	1.57	0.290	7.37	1.844	46.84
.0625	1.588	0.344	8.74	2.187	55.55
.075	1.91				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. There shall be affixed to the relay a legible circuit diagram that permanently and positively identifies each terminal location specified.
5. The use of diodes on ac relays is optional. The actual application must be shown on label.
6. Pins shall be perpendicular to the header surface within one degree.
7. In the event of conflict between the text of this standard and the references cited herein, the text off this standard shall take precedence.
8. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this specification sheet to the extent specified herein.

FIGURE 1. Dimensions and configuration – Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Load ratings: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 40 amperes dc; 60 amperes ac.

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Rupture current (NO): 50 amperes dc; 80 amperes ac.

Coil data: See table III.

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength:

Sea level, 2-5 seconds:

Coil to case:	Initial	After life tests
	1,000 V rms	750 V rms
Aux. contacts:	N/A	N/A
All other points:	1,000 V rms	750 V rms

Dielectric strength (altitude) (80,000 ft.):

Coil to case:	250 V rms	N/A
Aux. contacts:	N/A	N/A
All other points:	250 V rms	N/A

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock, g-level: 50 g's.

Duration: 6 ms.

Maximum duration contact opening: 100 µs.

Vibration, sinusoidal:

G-level: 10 g's.

Frequency range: 5-1500 Hz.

Acceleration: 15 g's.

Terminal strength (high temperature pull and torque test): Not applicable.

Part or Identifying Number (PIN): MS25323- (plus applicable dash number from table I).

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TABLE I. Dash numbers and general characteristics.

PIN MS25323-	Type	Coil type	Terminal type	Max weight (pounds)
D2	1	dc	Plug in	0.30
A2	1	ac	Plug in	0.32
A3	1	ac	Plug in	0.32

TABLE II. Rated contact load (amperes per pole) (case grounded). 1/

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 2/				See Appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	20	6	6			6	4			6	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														3/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Mixed loads	Applicable in accordance with MIL-PRF-6106													

1/ Absence of value indicates parameter is not applicable to this specification sheet.
2/ Absence of value indicates relay is not rated for 3-phase applications.
3/ Transfer load indicates that the relay is suitable for transfer between unsynchronized ac power supplies at the rating indicated.

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TABLE III. Operating characteristics.

PIN MS25323-	Coil	Coil data										Time (milliseconds maximum)					
		Rated			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Contact bounce 3/			
														Main		Aux 6/	
		Volts 1/	Freq. Hz	Res Ω	Volts	A	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
D2	X1,X2	28	dc	N/A	29	0.20	18	19.8	22.5	7.0	1.5	20	20	2	2		
A2	X1,X2	115	400 5/	N/A	122	0.07	90	95	103	35	5.0	25	50	2	2		
A3	X1,X2	115	50/60	N/A	122	0.10	90	95	103	35	5.0	25	50	2	2		

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.
5/ MS25323-A2 may be used on 60 Hz if maximum ambient temperature is limited to +85°C.
6/ Absence of value indicates parameter is not applicable.

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Quality conformance inspection:

Group B and group C testing are not required. In the event of a change in the design or construction of the part, the manufacturer shall notify the qualifying activity and shall impose additional testing requirements as necessary.

Group A acceptance reports shall be submitted to the qualifying activity every 36 months to retain qualification for this military standard sheet.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:
MIL-STD-461 SAE-AM3332 MIL-DTL-45204

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA-CC

(Project 5945-2024-056)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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